Teaching Philosophy — Connecting Engineering Theory, Experimentation, and Professional Practice

Waseem Raza, PhD | Kettering University | Electrical and Computer Engineering (FP00103)

Teaching philosophy and instructional experience

My teaching philosophy is that engineering knowledge becomes durable when students can explain a concept, implement it, test its limits, and defend their conclusions. A rigorous lecture establishes models and assumptions; simulation makes their consequences visible; laboratory and project work expose practical constraints; and reflection helps students reconcile predictions with observations. This progression is particularly important in electrical and computer engineering, where abstract mathematics must ultimately guide the behavior of real systems.

As a Lecturer at the University of Lahore, I taught undergraduate Computer Programming, Communication Systems, and Wireless Communication. I prepared lectures and assessments, supported student projects, and explained complex subjects through structured problems and implementation-oriented examples. My earlier appointment in the Communication Systems Lab at the University of Engineering and Technology, Taxila involved communications laboratory instruction, experiment preparation, assignments, and evaluation. At COMSATS University Islamabad, I mentored Bachelor's and Master's research and project students in experimental design, simulation, analysis, and technical reporting. These experiences taught me to balance intellectual rigor with clear expectations and accessible guidance.

From mathematical models to evidence

I design lessons around a progression: what the model predicts, how to compute the prediction, what we observe, and why the two may differ. In a communication systems course, students might begin with Fourier-domain reasoning and modulation theory, implement a Python or MATLAB simulation, examine performance under noise and channel distortion, and then explain discrepancies between theoretical and numerical results. A signals-and-systems exercise could compare analytical convolution with sampled signals, or investigate spectral leakage and filtering choices. Students would submit reproducible code, figures, and brief engineering explanations rather than only final numerical answers.

Kettering's 2026–2027 curriculum includes EE-430 Communication Systems, EE-340 Electromagnetic Wave Propagation, EE-336 Continuous-Time Signals and Systems, EE-338 Discrete-Time Signals and Systems, and EE-434 Digital Signal Processing. These topics connect naturally to my teaching, research, and industrial work. I would prioritize contributions consistent with demonstrated subject preparation, while coordinating with departmental faculty and laboratory staff on any broader foundational electrical-engineering teaching assignment. Course readiness includes reviewing the syllabus, prerequisite expectations, laboratory instruments, and applicable safety procedures, not simply assuming that expertise in one engineering area automatically transfers to another.

Experiential learning, laboratories, and cooperative education

I see laboratory learning as guided engineering inquiry rather than a checklist of steps. A well-designed experiment should have clear prerequisites, a testable hypothesis, identifiable sources of uncertainty, and a deliverable that demonstrates both technical competence and engineering judgment. Before class, students receive a short preparation task; during the lab, they follow safe procedures while collecting or simulating evidence; afterward, they compare results with expectations and recommend improvements. In programming-oriented labs, version control, unit tests, readable plots, and honest reporting of errors reinforce professional habits.

Kettering's cooperative-education model presents a distinctive opportunity to connect classroom work with authentic engineering constraints. I would ask students to draw on co-op insights where confidentiality permits, such as why measurements need calibration, how design assumptions affect reliability, how teams document faults, or why software and hardware interfaces require careful validation. Students without directly relevant workplace assignments would be

given equivalent public or instructor-designed case studies, so access to a particular employer does not determine their opportunity to succeed.

Student projects can span multiple scales: introductory computational experiments, communication-link analysis, network measurement studies, sensing demonstrations, and interdisciplinary design challenges. My experience with AT&T and Futurewei provides real engineering examples involving network diagnostics, signal analysis, data quality, and reproducible evaluation. I would also support capstones by helping teams define measurable requirements, manage milestones, assess safety and constraints, and communicate results professionally.

Assessment, inclusion, and continuous improvement

Students learn best when the criteria for success are explicit and feedback arrives in time to influence their next attempt. I use a combination of short diagnostic questions, scaffolded assignments, laboratory observations, project milestones, and summative assessments. Rubrics would distinguish conceptual reasoning, correctness, verification, reproducibility, and communication. An incorrect answer with a clear and testable method is an opportunity for targeted feedback; an apparently correct answer with unsupported assumptions still warrants scrutiny. For team projects, individual contributions and reflections help make evaluation fair.

Inclusive teaching begins with clear explanations and multiple pathways into the same technical idea: equations, diagrams, computational experiments, and physical intuition. I would provide predictable office hours, accessible materials, opportunities for low-stakes practice, and constructive feedback while maintaining rigorous standards. I emphasize academic integrity, appropriate use of AI-assisted coding tools, transparency about sources, and independent understanding of submitted work.

Finally, I regard teaching as an iterative engineering process. I would review outcome-level assessment evidence, student feedback, and recurring problem areas with colleagues, then revise examples, pacing, and lab instructions accordingly. I would contribute to curriculum development and program assessment where needed, and engage professional partners in refining projects to reflect emerging technology without sacrificing foundational knowledge.

My goal at Kettering is to help students move confidently between theory, laboratory evidence, and industrial practice, developing the analytical habits, professional responsibility, and curiosity needed for meaningful engineering careers.